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A novel method for computing the 3D friction cone using complimentary constraints

Dean Pretorius[1], *Student Member, IEEE,* Callen Fisher[1], *Member, IEEE*

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***Abstract*— Modeling the Coulomb Friction Cone in trajectory optimization is typically done by linearizing it along a series of vectors. Often, these vectors define the edges of polyhedral estimations of the cone. This article provides an alternate approach that samples the cone along a vector that satisfies the Maximum Dissipation Principle, which is shown to be significantly more computationally tractable. The proposed technique uses the polar representation of the relative velocity of a contact point on a surface to determine the direction of the resultant friction force and linearizes the friction cone along this vector. This study describes the development of the proposed model. Thereafter, a trajectory optimization experiment was conducted to compare the traditional four-sided polyhedral estimation of the friction cone with the novel method. Compute time and computational complexity were used as performance metrics in this study. Results from these experiments indicate that the proposed method reduced the compute time by 39.13% and a 57.00% reduction in inequality constraints when compared to the 4-sided polyhedral estimation.**

I. Introduction

Legged robots must effectively leverage contact events to orient, balance, and move themselves in 3D space. Achieving this will free robots from their support structures to explore and do useful work beyond the controlled confines of the laboratory [1].

Therefore, roboticists must develop a comprehensive understanding of intermittent and non-planar contact models. However, intermittent ground contact events are inherently discontinuous and impulsive; further developing their notoriety for being computationally intractable [2], [3].

Consequently, work done by Patel, Stewart, and Posa et al. [2]–[6] use *contact-implicit* trajectory optimization methods to better study the discontinuous mechanics inherent in effective locomotion. These methods allow the optimizer to make and break contact, while accounting for the effects of friction, and slip [7].

Trajectory optimization is fundamental to these methods. It is a computational tool used to find solutions to nonlinear problems, while satisfying a set of constraints, and minimizing a cost function [7], [8]. In these optimizations, inelastic contact events, friction, and slip are accurately modeled

[1] Authors are with the Department of Electrical Engineering, Stellenbosch University

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[1]Dean Pretorius is with the Department of Electrical and Electronic Engineering, University of Stellenbosch, 7599 Stellenbosch, South Africa19934084@sun.ac.za

[2]Callen Fisher is with the Department of Electrical and Electronic Engineering,University of Stellenbosch, 7599 Stellenbosch, South Africa cfisher@sun.ac.za

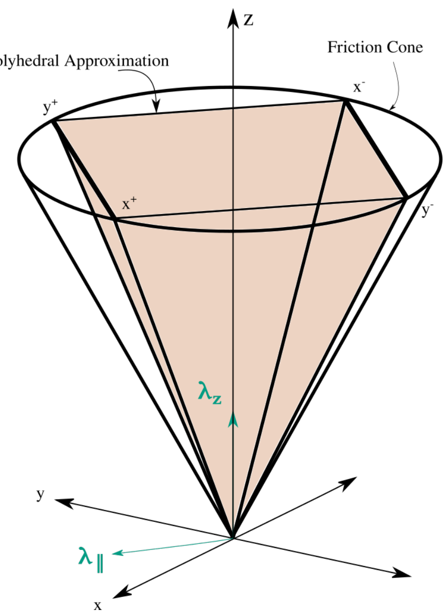

Fig. 1. A graphic representation of the four-sided polyhedral approximation of the 3D friction cone [4]. The edges of the friction cone are defined along the positive and negative directions of the Cartesian axes.

as a set of Mathematical Programs with Complimentarity Constraints (MPCC) [2], [5]. However, these solutions are often constrained to planar models due to their complexity.

Extending these methods to include non-planar models requires an accurate model of 3D friction. Friction is described by the Maximum Dissipation Principle as a force that maximizes the rate of kinetic energy dissipation at contact [6], [9], [10]. On a surface with a uniform coefficient of friction, μ, the resultant friction force is colinear to, and opposes, the direction of relative velocity. Since this constraint is applied symmetrically around the normal contact force, λ_z, it is visualized as an isometric cone around the point of contact. Adhering to Coulomb's law requires the magnitude of the horizontal component of the ground reaction force, $\boldsymbol{\lambda}_{\|} = [\lambda_x, \lambda_y]^T$, to always be less than or equal to a proportional value of the vertical component of the ground reaction force, λ_z, such that $|\boldsymbol{\lambda}_{\|}| \leq \mu\lambda_z$ [4], [9]–[12].

Furthermore, modeling the friction cone using MPCC techniques require that it be linearized. Therefore, solutions are found by replacing the friction cone with an inner polyhedral approximation of the cone [4] [2]. These approximations are done using a series of linear constraints that find solutions along the edges of the polyhedral [2], [4], [6]. In a planar environment the edge of the polyhedral lies on the plane, resulting in an accurate model of the resultant friction forces, while being efficient in terms of computational cost. However, in a 3D environment the edges of the polyhedral approximation are defined by Cartesian vectors and the resultant friction force is split across these vectors, approximating the cone as shown in Fig. 1 [4].

This approach is computationally expensive and results in inaccuracies. When the direction of relative motion between the point of contact and the surface is not on, or close to, the edges that define the approximation, a kinetically feasible underestimation of the resultant friction forces emerge. Additionally, this underestimation is maximized when the direction of relative motion lies in between the edges of the approximation. Nevertheless, increasing the amount of edges defining the polyhedral approximation increases the the accuracy of the approximation, while also increasing the computational complexity of the model [2], [4].

This paper proposes a novel method that leverages the Maximum Dissipation Principle's dependence on the relative velocity of a contact event to accurately and efficiently linearize the 3D friction cone in non-planar trajectory optimization models. Employing the Maximum Dissipation Principle only finds solutions along the vector that describes the direction of the relative velocity of the contact point and the surface, which is colinear to the direction of resultant friction. This significantly reduces the amount of constraints needed to model the 3D friction cone, and the computation time needed to solve these contact implicit optimization models. Therefore, addressing the inaccuracies and computational inefficiencies present in the polyhedral approximation of the 3D friction cone.

An experiment was conducted in which both the proposed method as well as the commonly used four-sided polyhedral approximation of the friction cone were implemented in trajectory optimization and compared [4], [13], [14]. Performance was measured in terms of compute time averaged over 20 experiments and computational complexity compared the amount of variables and constraints required for each respective optimization. Results from the experiments discussed in this paper indicated that the proposed method reduced the compute time by 39.13% and required 57.00% fewer inequality constraints when compared to the polyhedral estimation.

This article begins by describing the development of the trajectory optimization models in Section II. Thereafter, Section III describes the outcomes of the experiments conducted in this paper. These results are analyzed and discussed in Section IV. The paper concludes with Section V.

II. Method

This paper compares our novel method of modeling the isometric friction cone, with the the polyhedral approximation of the friction cone, in non-planar trajectory optimization models [2], [4], [6]. An experiment was conducted in which two trajectory optimizations of a monopod robot, each implementing their respective contact model, running at an angle of $45°$ was optimized. This method included a 3D free-bodied robot, interacting with a 3D coordinate frame, as shown in Fig. 2. The computation time taken to find feasible solutions, as well as the computational complexity, of these non-planar models were used as metrics to compare the two models. A detailed description of the development of the trajectory optimization models are provided below:

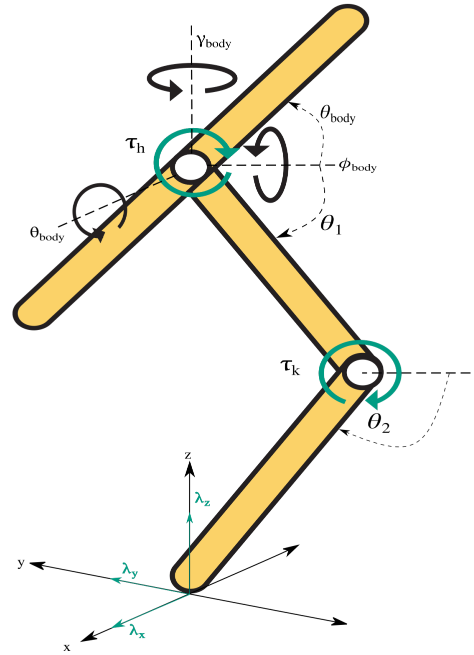

Fig. 2. A monopod robot was modeled in this study. It was made of an 3-link rigid body system, consisting of a double pendulum and a body link. Motor torques only actuate in the θ direction, and external ground reaction forces are present.

A. Dynamics

Euler-Lagrange dynamics, in the form of the manipulator equation, were used to describe the equations of motion of the system [15],

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{B}\boldsymbol{\tau} + \mathbf{A}\boldsymbol{\lambda} \quad (1)$$

where $\mathbf{M}(\mathbf{q})$ was the mass matrix, $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})$ described Coriolis forces, and $\mathbf{G}(\mathbf{q})$ described the gravitational potentials. $\mathbf{q}$ represented the generalized coordinates of the monopod robot, $\mathbf{q} = [x, y, z, \phi_{body}, \theta_{body}, \gamma_{body}, \theta_1, \theta_2]^T$, as shown in Fig. 2.

Euler Angles (ϕ_{body}, θ_{body}, γ_{body}), expressed relative to the inertial frame, were used to describe 3D orientation of each rigid body [16]. Thus, reducing the computational complexity of the model by simplifying the equations of motion and improving the sparsity of the Coriolis term in (1) [17].

Additionally, $\mathbf{A}$ and $\mathbf{B}$ mapped the motor torques, for the hip and knee, and external ground reaction forces, $\boldsymbol{\tau} = [\tau_h, \tau_k]^T$ and $\boldsymbol{\lambda} = [\lambda_x, \lambda_y, \lambda_z]^T$ respectively, to the relevant generalized coordinates, $\mathbf{q}$.

Sympy [18] was used to develop the linear algebra that describes these dynamics. These dynamics were interpreted as constraints describing the equations of motion for the respective state variable, allowing the optimization to find a feasible solution to the equations of motion.

B. Model Parameters

Table I describes the parameters of the monopod shown in Fig. 2. Each link was modeled as a solid cylindrical tube rotating around the inertia axes, resulting in the center of mass for each link being at the center of the link. All links had uniformed radii, $2.5cm$, and length, $1.0m$. Whereas, the body mass, $m_{body} = 0.75kg$, differed from the mass of the leg links, $m_{link} = 2.0kg$.

TABLE I
PARAMETERS OF MONOPOD

Parameter	Symbol	Value
Radius of links	r	2.5 cm
Length of links	l	1.0 m
Mass of leg links	m_{link}	2.0 kg
Mass of body link	m_{body}	0.75 kg
Coefficient of friction	μ	0.8

C. Constraints

During contact events, the dynamics described in Section II-A are highly nonlinear. The relevant constraints that helped the solver find kinematically feasible solutions are described below.

1) Discretization: All trajectories generated in this experiment were discretized into N=50 node points. Due to intermittent contact events, these trajectories were highly impulsive and nonlinear. Therefore, a variable time-step was employed to increase the resolution during periods of high non-linearity, and decrease the resolution when the trajectory can be accurately described using linear functions.

The following bounds were placed on this variable time-step:

$$0.1h_M \leq h_i \leq 2h_M \tag{2}$$

where h_i was the duration of the i^{th} node, such that $i = 1, 2, ..., N$, and $h_M = \frac{T}{N}$ described a ratio between an estimated time needed for the optimizer to complete its tasks, $T = 0.5s$, and N.

Additionally, equations of motion were further discretized using K-point Runge-Kutta collocation to increase the accuracy of the optimization by an order of h^{2K-1}. As adapted from Patel et al. [3] third order polynomials, K=3, were used to join the collocation points.

2) Mathematical Programs of Complimentarity Constriants (MPCC): Contact-implicit optimization methods use MPCCs to allow the optimizer to determine kinematically feasible footfall patterns for a given trajectory. All contact events were described as inelastic impacts, and friction was bounded by Coulomb's law [10], [12].

This contact model can be found in [2], equations (8) to (16). However, the MPCCs were been adapted such that ϵ-relaxation methods help the solver find solutions for these constraints [19], [20] and three-point collocation methods increase its accuracy [3].

All MPCCs were implemented as follows:

$$\begin{gathered} \alpha_{ij} \geq 0, \qquad \beta_{ij} \geq, 0 \\ \alpha_i' = \sum_{j=0}^{K} \alpha_{ij}, \qquad \beta_i' = \sum_{j=0}^{K} \beta_{ij}, \\ \alpha_i' \beta_i' \leq \epsilon, \end{gathered} \tag{3}$$

where α and β were both discretized to the i^{th} node and the j^{th} collocation point. These variables were summed across the collocation points, j, and the MPCC was only evaluated at the node points, i [3].

In applying the ϵ-relaxation methods, MPCCs were framed as inequality constraints with an upper bound. This relaxation parameter was initially set high and the model was solved iteratively, each time reducing ϵ until a desired accuracy was achieved [1].

3) Polyhedral Approximation of friction cone: Posa et al. [2] provides a contact model that implements a polyhedral approximation of the friction cone in [2], equations (30) to (34). This approximation of the friction cone uses unit vectors $\mathbf{D}^k$, for $k = 1, 2, ..., d$, to define the edges of this polyhedral. In addition, the net friction force is the sum of the magnitudes along the components making up the friction cone, such that $|\boldsymbol{\lambda}_{\parallel}| = \sum_{k=0}^{d} \mathbf{D}^k \lambda^k$, where λ^k is a scalar that describes the magnitude of the friction force in the direction of $\mathbf{D}^k$.

In this experiment the polyhedral approximation was modeled as a four-sided pyramid, such that $d = 4$, shown in Fig. 1 [4], [13], [14]. All references to the polyhedral approximation of the friction cone in this paper refers to the four-sided approximation of the friction cone used in the experiment. Therefore, $\mathbf{D}$ described unit vectors along the positive and negative components of the x and y axes, where $\mathbf{D} = [x^+, x^-, y^+, y^-]^T$. The signage of the superscript denoted the direction of the unit vector along the respective axis [2], [4], [6].

4) 3D Friction Cone: As an alternative to Section II-C.3, the method proposed in this paper leverages The Maximum Dissipation Principle's dependence on relative velocity to determine the direction of the resultant friction [10]. In addition, contact-implicit optimization techniques [2] [4] were used to compute the magnitude of the resultant friction, while adhering to Coulomb's Law.

The proposed method models contact forces as, $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_{\parallel}, \lambda_z]^T$, such that λ_z is normal to the contact surface and $\boldsymbol{\lambda}_{\parallel}$ represents the tangential component of resultant friction, where $\boldsymbol{\lambda}_{\parallel} = [\lambda_x, \lambda_y]$. $\boldsymbol{v_{rel}} = [v_x, v_y]$ represents the relative velocity at the point of contact with respect to ground and is mapped to the generalized coordinates, $\mathbf{q}$. These Cartesian vectors are interpreted in polar form as shown in Fig. 3 and (4).

When discretized, the method proposed in this paper is described as,

$$\begin{gathered} \boldsymbol{\lambda}_{i,\parallel} = |\boldsymbol{\lambda}_{i,\parallel}| \angle \Theta_i \\ \boldsymbol{v}_{i,rel} = |\boldsymbol{v}_{i,rel}| \angle - \Theta_i. \end{gathered} \tag{4}$$

[1] This is further described in Section II-G

Θ is an auxiliary variable that represents the direction of the tangential component of friction with respect to the x-axis. Consequently, the optimizer finds a series of Θ values that directly opposes the direction of $\boldsymbol{v_{rel}}$ [10].

The contact model is described as:

$$\rho(\mathbf{q}_{i+1}) \geq 0, \lambda_{i,z} \geq 0 \tag{5}$$

$$\rho(\mathbf{q}_{i+1})\lambda_{i,z} = 0 \tag{6}$$

$$|\boldsymbol{v}_{i,rel}|(\mu\lambda_{i,z} - |\lambda_{i,\parallel}|) = 0 \tag{7}$$

$$|\boldsymbol{v}_{i,rel}||\boldsymbol{\lambda}_{i,\parallel}| = 0 \tag{8}$$

where $\rho(\mathbf{q})$ represents the distance between the foot and ground[2]. (5) and (6) represent the non-penetration constraint that describes inelastic contact events [2], [3]. Similarly, (7) and (8) applies the friction cone. This ensures that, if slip occurs, the magnitude of the friction force is bounded by the edge of the friction cone. Inverse kinematics determines the magnitude of the resultant slip, while satisfying (1). Thereafter, the polar form of $\boldsymbol{\lambda}_{i,\parallel}$ is decomposed into its Cartesian elements such that

$$\lambda_{i,x} = |\lambda_{i,\parallel}|cos(\Theta_i) \tag{9}$$

$$\lambda_{i,y} = |\lambda_{i,\parallel}|sin(\Theta_i). \tag{10}$$

Collectively, (4) to (10) describe a discretized contact model that accurately computes the resultant components of friction in 3D.

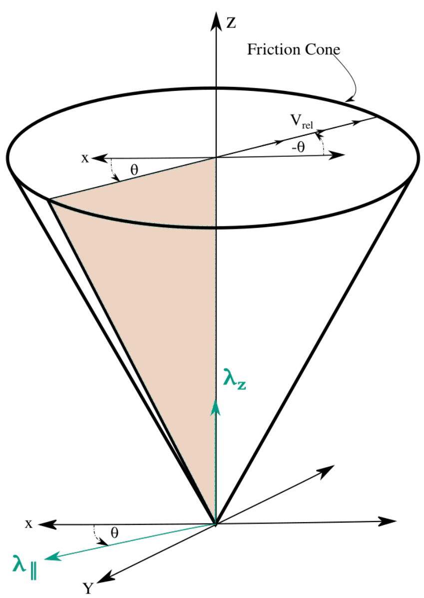

Fig. 3. A visual explanation of the proposed method of modeling the 3D friction cone. The resultant friction force directly opposes the direction of relative velocity.

D. Bounds on motion

Variables were restricted to reduce the search space of the optimization while still allowing the solver to find kinetically feasible solutions. These restrictions were implemented using a combination of bounds and constraints [7].

Constraints were used to place limits on $\mathbf{q}$, $\dot{\mathbf{q}}$ to enforce a feasible range of motion. Each component describing the absolute position of the robot (x, y, z) was positively bound between the origin, and $1.0m$ between the origin

For the robot to maintain an upright posture, the roll of the body, ϕ_{body}, was restricted between $-10°$ and $10°$. Whereas, the yaw of the body, γ_{body}, was restricted between $-90°$ and $0°$ to allow the monopod to move away from both the x and y axes[3].

Since joint angles were defined using absolute angles, constraints were placed on the respective angle relative to the pitch, θ_{body}, as shown:

$$\begin{gathered} -90° < \theta_{\text{body}} < 90°, \\ 0° < \theta_1 - \theta_{\text{body}} < 180°, \\ 0° < \theta_2 - \theta_1 < 110°. \end{gathered} \tag{11}$$

Similar constraints were placed on the resulting angular rates.

All variables used in MPCC constraints were bound positively and all angles describing the direction of force and velocity vectors were bound between $-180°$ and $180°$. Actuator torque was bound between $50N.m$ positively and negatively to prevent the input torque actuators contributing unrealistic amounts of energy to the simulation.

E. Cost Function

A minimum torque cost function further restricted the input motor torque within the bounds described in Section II-D. Actuator torque was only computed at node points, resulting in the following cost function:

$$J = \sum_{i=0}^{N} \tau_{i,h}^2 + \tau_{i,k}^2 \tag{12}$$

where $\tau_{i,h}$ and $\tau_{i,k}$ represented the actuator torques discretized at the i^{th} node of the hip and the knee respectively.

F. Initial and terminal conditions

Initial conditions were set to start the robot at the apex of a hop, at the origin of the inertial plane. While terminal conditions were constrained to force motion away from the x and y axes that define the polyhedral approximation of the friction cone.

To start the simulation at the apex of a hop, the vertical velocity and acceleration of the first node was set such that $\dot{z}_0 = 0.0\frac{m}{s}$ and $\ddot{z}_0 = -g\frac{m}{s^2}$ where g is the gravitational constant. The model was set to start at the origin of the Cartesian plane, such that $x_0 = 0.0m$ and $y_0 = 0.0m$. To move the robot away from the axes that defined the polyhedral approximation of the friction cone final conditions were constrained, such that $x_{N-1} = y_{N-1}$. The subscript 0 denotes the first element, and $N-1$ denotes the final element, of the discretized variable.

[2]Previous literature often represents the distance between the foot and the ground as $\phi(\mathbf{q})$ [2], [3], [6], [21]. However, this notation conflicts with the notation used to represent the roll, ϕ_{body}, orientation angle described in Section II-A.

[3]This is done to maximize the error of the polyhedral estimation as described in II-G

G. Solver Set-up and Seed Generation

The nonlinearities of these optimization problems made it easy for the solver to get stuck in deep local minima. This made finding feasible trajectories computationally intractable. However, certain methods were used to make it easier for the solver to find solutions.

At initialization, all generalized coordinates were uniformly randomized within their bounds, and the rest of the state variables were set to 0.01 [7]. Both optimizations were initialized with the same starting seed to prevent a bad starting seed affecting the performance of an individual optimization model [17].

Thereafter, the cost function was set to 1 and variable tolerances were increased, for the initial solve to obtain a more feasible seed [3]. Once warm started, variable tolerances were decreased, the minimum energy cost function was activated, and the iterative ϵ-relaxation techniques were used to find an optimal solution, as described in Section II-C.2. Initially, ϵ was set to 1000. Thereafter, the optimization was solved iteratively, with each iteration dividing ϵ by 10 until $\epsilon = 0.0001$ [3], [7]. After 8 successful iterative solutions, the MPCCs were satisfied with $\epsilon = 0.0001$ for all complimentarity constraints and the solution was deemed feasible.

III. Experiments and Results

The aim of the experiment was to compare the performance of a contact model that implemented our novel method for computing the friction cone, and a contact model that implemented the polyhedral approximation of the friction cone which is commonly found in the literature [2], [4], [6], [13]. A trajectory optimization model of a 3D, free-bodied, monopod robot as described in Section II was implemented for each contact model. For both contact models, periodic steady-state motion was achieved; thereby, achieving effective hopping.

The trajectory optimization methods described in this paper were implemented in the Python optimization package, Pyomo [22], [23]. The IPOPT solver [24] was used to find solutions for the optimization models described in this paper. All computation was completed in the Google Colab environment which provides 12Gb of RAM to run the experiments on two 2.3GHz Intel Xeon CPUs [25]. A number of observations and comparisons were made and are shown below.

A. Comparison of Compute Time

To compare the time needed to compute the respective methods for modeling the friction cone, 20 experiments were conducted and the time needed for the optimizer to find a feasible solution was recorded. It should be noted that both optimization models were initialized with the same randomized seed. Therefore ensuring that both optimizations were compared using the same reference point.

Fig. 4 shows time needed to solve a model implementing the respective model of the friction cone. It is evident that the novel method of modeling the friction cone computes faster than the traditional polyhedral estimation of the friction cone. On average, a solution for our novel method was found 39.13% faster than a solution for the polyhedral estimation. Further discussion is provided in Section IV.

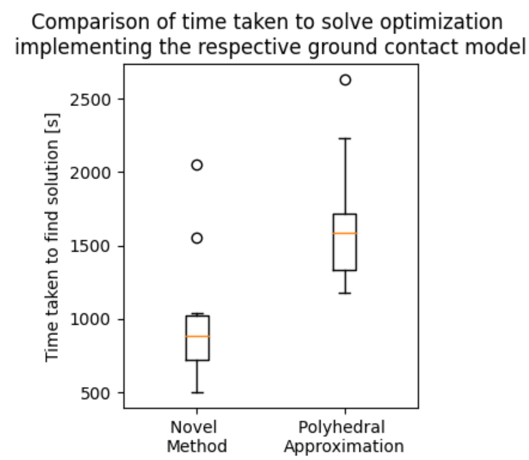

Fig. 4. A plot showing the time taken to find optimal solutions for trajectory optimization models applying the respective methods across 20 experiments. The orange line represents the median, and the inter-quartile range is bounded by the box.

B. Comparison of Computational Complexity

Computational complexity was judged by comparing the amount of variables and equations that the optimizer passes to the solver. Table II compares the amount variables, and constraints generated in the model and passed to the solver. The results shown in Table II are extracts from the information displayed before the solver started up before each iterative solve. It is noted that the novel method passed 450 fewer variables, 300 more equality constraints, and 200 fewer upper bound inequality constraints. Further discussion is provided in Section IV

TABLE II

A tally of variables, and constraints passed to the IPOPT Solver for the trajectory optimization method implementing the respective method

	Novel Method	Polyhedral Approximation
Number of variables	6101	6551
Number of equality constraints	5648	5348
Upper bound inequality constraints	149	349

C. Optimal Trajectories

Fig. 5 provides a set of graphs from optimal trajectories of models that implemented both methods. Animations of these trajectories are available in the supplementary video. Contact occured between nodes, 15 and 35. Therefore the magnitude of friction, $|\lambda_{\|}|$, and effective coefficient of friction, $\mu = \lambda_{\|}/\lambda_z$, was determined. An impulse was noticed on the |friction| graph, in Fig. 5, as the foot initially made contact with the ground. Thereafter, the magnitude of friction settled at a value proportional to the weight of the robot.

Although the optimizations were initialized with the same randomized seed, they were different optimization models, described by different equations, with different gradients in the solution space. This resulted in the optimizer finding different solutions to the respective optimization. No slip occurred in these trajectories as their effective μ remained less than the coefficient of friction set for this experiment, $\mu = 0.8$ [13].

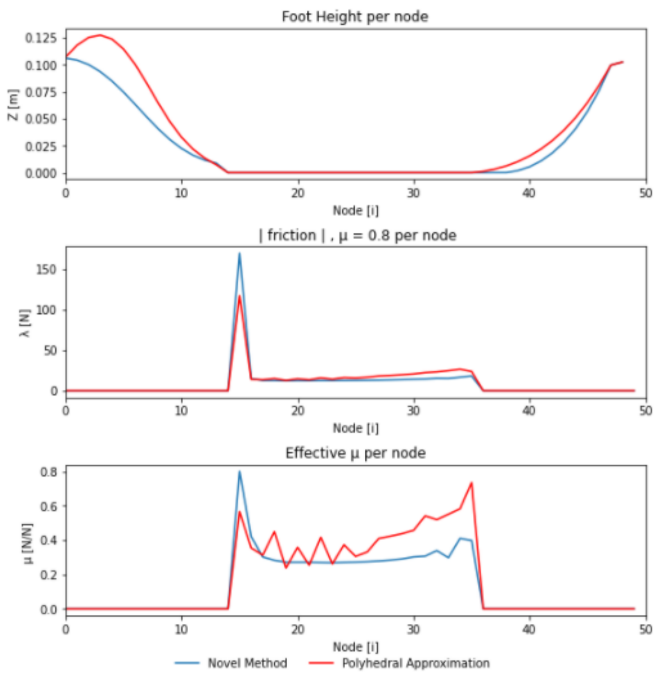

Fig. 5. A set of optimal results are provided displaying three graphs relevant to the trajectory. The blue line represents the respective graph of the optimization containing the novel contact model that implemented the novel method of modeling the friction cone. Similarly, the red lines represents the respective graph relating to the trajectory of the optimization containing the polyhedral approximation of the friction cone. The first graph in 5 represents the trajectory of the foot height, $\rho(q)$, through the simulation. This is followed by a graph representing the magnitude of the resultant horizontal friction force, $|\boldsymbol{\lambda}_{||}|$. Thereafter, a graph of the effective coefficient of friction, $|\boldsymbol{\lambda}_{||}|/\lambda_z$, is provided.

IV. Discussion

The results described in Section III are significant as they indicate the benefits of using the novel method for computing the friction cone in 3D. It was shown that the novel method is more computationally efficient, and solves faster.

On average, it took 994s for the optimizer to find a solution for the model containing the proposed method of computing friction. Whereas, it took the optimizer an average of 1633s to find a solution for the model containing a four-sided approximation of the friction cone. This indicates that the novel method of friction modeling, takes 39.13% less time to solve.

ϵ-relaxation methods framed the MPCC problems as inequality constraints with an upper bound. It is noted that the implementation of the polyhedral approximation of the friction cone passed 349 upper bound inequality constraints to the solver, while the implementation of the novel method passed 149 as seen in Table II.

As a result, the novel method of modeling the friction cone reduced the amount of MPCC problems in the optimization by 57.00% when compared to the four-sided polyhedral approximation of the friction cone.

Implementations of the proposed method applies and MPCC only along the vector that opposes the direction of foot velocity. In contrast, implementations of the polyhedral approximation of the friction cone applies an MPCC along each vector that defines an edge of the approximation.

Therefore, increasing the amount of faces of the polyhedral estimation increases the accuracy of the estimation of the friction cone. However, it increases the amount of MPCCs present in the optimization problem - making it more computationally intractable.

In summary, the novel method of computing the friction cone was noticeably more computationally efficient and required significantly less compute time to find a solution than the traditional four-sided polyhedral approximation of the friction cone.

V. Conclusion and Future Work

This article presented a new method of modeling the 3D friction cone in contact-implicit trajectory optimization. Additionally, experiments were conducted to prove the working of this model, and to compare it against existing methods of modeling the 3D friction cone.

Previous methods of approximating the friction cone are notorious for being computationally inefficient, complex, and inaccurate. However, results from this study indicated that the proposed method addresses these limitations by being significantly more computationally efficient and tractable.

Moreover, this work develops the field of contact-implicit optimization by adding to the findings of [2]–[4], [7]. Knowledge gained from this study will help roboticists better understand contact events in non-planar environments. Future research involves implementing the proposed method on multi-legged bipedal, and quadrupedal, robots to investigate the scalability of the proposed method.

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